A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Properties of pollutants & environmental factors - Set 1

Mark scheme

AQA spec 3.4.1-3.4.2; Pollution; textbook pp. 236-242

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. **1** The time taken for half of a pollutant (released into the environment) to break down / degrade **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	1

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** DDT is liposoluble so it is stored in fat / can pass through phospholipid cell membranes and is not excreted (nitrates are water soluble and excreted easily) **1** DDT is persistent (long half-life) so it is not broken down before it is passed to the next trophic level **1** So DDT bioaccumulates in each organism and becomes more concentrated at each successive trophic level **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time • **half-life** → the time for the activity, or the amount, to fall by half • **trophic level** → the feeding level in a food chain **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
01.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** The gas is denser than air so it settled close to the ground / did not rise and disperse **1** Low wind velocity (at night) meant it was not dispersed / diluted **1** Short half-life so it broke down before travelling far **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** High solubility means nitrates are easily dispersed / leached / carried by water so they affect a larger area (eg rivers, aquifers, the sea) **1** But dispersal dilutes the pollutant so the concentration at any one point may be lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water	2

Do not write outside the box

Question	Marking guidance	Marks
01.5	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Noise / sound **1** Heat / thermal pollution **1** Ionising radiation **1** Light **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Ionising radiation** → radiation with enough energy to remove electrons and form ions **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1 Mutagen:** an agent that changes the chemical structure / base sequence of DNA (causes mutations), eg ionising radiation, UV, PCBs, cadmium, asbestos **1 Carcinogen:** a mutagen that causes cancer / uncontrolled cell division producing a tumour **1 Teratogen:** an agent that causes birth abnormalities by preventing normal gene expression (inhibiting proteins / enzymes) without changing DNA structure, eg mercury, 2,4,5-T **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **ionising radiation** → radiation with enough energy to remove electrons and form ions **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Teratogens do not change the DNA structure, so the abnormality is not in the genes and cannot be passed on **1** Mutagens change DNA – if the mutation is in a gamete / egg / sperm (gonadic effect) it can be inherited **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1 Somatic:** mutation in a general body cell, affecting the individual only (eg cancer / tumour) **1 Gonadic:** mutation in cells of the ovaries or testes / gametes / embryo, which may cause abnormalities in offspring **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.4	**1 mark each** 1 mark for each of the 3 points below. **1** Two or more pollutants whose effects interact to produce a different (usually more serious) effect **1** The pollutants themselves do not react to form a new pollutant (that would be a secondary pollutant) **1 Example:** ozone damages leaf cuticles so sulfur dioxide can cause more damage to the exposed cells – smoke and sulfur dioxide in smogs – neonicotinoids with fungicides **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **synergism** → two pollutants together doing more damage than the sum of each alone	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 239-240; AQA spec 3.4.1.	
03.1	1 mark for the point below. **1** Temperature inversion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Temperature inversion** → a layer of warm air above cooler air that traps pollutants	1
03.2	**1 mark each** 1 mark for each of the 4 points below. **1** On night A, temperature falls with altitude – warm effluent gases are less dense / more buoyant than the surrounding air so they rise **1** As they rise they cool but the surrounding air is also cooler so they stay buoyant and continue to rise, dispersing and diluting **1** On night B the air near the ground is colder / denser than the air above (inversion layer at about 400 m) **1** Pollutants cool as they rise and become denser than the warmer air above so they are trapped below the inversion layer and become concentrated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Cloudless sky allows infrared energy to be radiated from the ground so the ground cools rapidly **1** The cold ground cools the air in contact with it, producing a cold dense layer near the ground **1** Low wind velocity means the air layers of different temperatures do not mix **1** Valley allows cold dense air to collect / drain into the low ground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	3
03.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Smoke smogs (eg London 1952) **1** Photochemical smogs (eg Los Angeles, Mexico City) **1** Build-up of NOx / carbon monoxide / PM10 in cities **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **PM10** → fine solid particles / smoke and dust **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 241-242; AQA spec 3.4.2.2.	
04.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Temperature:** most chemical / biological reactions are faster at higher temperature so degradable pollutants (eg sewage) break down faster – but faster decomposition may increase deoxygenation **2 Light:** provides the activation energy for chemical reactions – photodegradation of pesticides (eg neonicotinoids break down fastest in sunlight) – light drives photochemical smog reactions producing secondary pollutants **2 Oxygen:** needed for aerobic bacterial decomposition of organic wastes / sewage – oxidation of sulfide ores produces sulfur dioxide / acid mine drainage – oxidation state affects the solubility of metals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **deoxygenation** → loss of dissolved oxygen from the water • **photochemical smog** → ground-level ozone and PANs formed by sunlight acting on oxides of nitrogen and VOCs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • ground-level ozone described as protecting us from UV - that is stratospheric ozone; at ground level it is a pollutant	6

Do not write outside the box

Question	Marking guidance	Marks
04.2	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Pollutant ions adsorb onto (the surface of) clay particles / organic matter in soil or sediment **1** So they are immobilised and cannot be absorbed by organisms / leached into water – radiation is absorbed in the ground **1** But they may be released later if conditions change (eg storms disturbing sediment, low pH) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Many heavy metals (eg lead, aluminium) are more soluble at low pH **1** So they are mobilised / leached from soil or sediments and can be absorbed by organisms / enter water bodies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 240-242, 255; AQA spec 3.4.2.2.	
05.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (4.8 - 1.4) / 4.8 x 100 **1** = 70.8% (accept 71%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** A tributary joins the river, more than doubling the flow (15 to 36 m^3 s^{-1}) **1** So the pollutant is diluted by the extra water (dispersal), not broken down **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** The pollutant is persistent so it is not degraded / broken down as it travels **1** Little extra water joins the river (flow 36 to 38) so little further dilution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	2
05.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Lake has a long residence time / water does not flow away so the pollutant accumulates rather than being carried away **1** Smaller volume of water / less dilution / less dispersal by currents **1** Persistent pollutant gradually becomes more concentrated (eg by evaporation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime • **Persistent** → does not break down / stays in the environment for a long time	2
05.5	**1 mark each** 1 mark for each of the 2 points below. **1 Marine discharge:** large volume of water and currents dilute and disperse the pollutant to acceptably low concentrations (control of emission location) **1 River:** small volume so concentrations stay high / drinking water abstraction / sensitive wildlife downstream **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abstraction** → water taken out of a river, lake or aquifer	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-241, 254-255; AQA spec 3.4.1, 3.4.2.1.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Tropospheric ozone (formed by reactions of NOx in sunlight)	1

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Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 4 points below. **1 Primary pollutant:** released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores – NOx from high-temperature combustion **1 Secondary pollutant:** produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) **1** Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid – NOx dissolve to form nitric acid **1** The acids (sulfurous, sulfuric, nitric) are secondary pollutants – ozone itself is a secondary pollutant of NOx photochemistry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	

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Question	Marking guidance	Marks
07.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **State of matter:** solids deposit near the source, gases disperse widely in the atmosphere – density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel • **Persistence / half-life:** persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution – non-persistent pollutants (sewage, organophosphates) can be allowed to degrade • Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk – LD50 / MDAF / MDNF used to set doses • Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) • **Reactivity:** reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants • Adsorption onto clay / sediments immobilises metals and caesium – can be used deliberately (activated carbon filters, polymer adsorption) • Mutagenic / carcinogenic / teratogenic action means no safe dose – exposure kept As Low As Reasonably Achievable (ALARA) • Synergism (ozone + SO2; smoke + SO2) means pollutants must be considered together • Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring • **Evaluation:** properties of new chemicals can be predicted from similar ones (precautionary principle)	9

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Question	Marking guidance	Marks
	– understanding allows prevention rather than clean-up **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **Persistence** → does not break down / stays in the environment for a long time • **half-life** → the time for the activity, or the amount, to fall by half • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	
	Model answer - Level 3 (9 marks)	
	A pollutant's properties largely determine where it goes and how much harm it does, so they can be used both to predict its behaviour and to choose a control. State of matter and density govern dispersal: solids deposit near the source, gases spread widely, and dense gases such as hydrogen cyanide at Bhopal hug the ground. Persistence decides whether dilution is enough: non-persistent wastes like sewage can be discharged where they break down, but organochlorines, CFCs, heavy metals and long-lived radioisotopes travel far and accumulate, so they need containment, bans or restrictions instead. Toxicity, specificity and mode of action, quantified by LD50 and used to set acceptable doses, identify which species are at risk. Solubility explains why nitrate leaches and is controlled by nitrate vulnerable zones, while liposolubility explains why DDT and methyl mercury bioaccumulate and biomagnify along food chains. Reactive pollutants may degrade quickly or form secondary pollutants - ozone, acid rain, PANs - so controlling the primary emissions with catalytic converters or flue-gas desulfurisation prevents the secondary ones. Adsorption on clay immobilises metals and caesium and is exploited deliberately in activated-carbon and polymer filters. Mutagens and carcinogens have no safe dose, so exposure is kept As Low As Reasonably Achievable, and synergism between ozone and sulfur dioxide, or smoke and sulfur dioxide, means pollutants must be assessed together.	
	Properties interact with the environment - inversions, currents, pH, oxygen - which is why Critical Pathway Analysis combines both to predict where pollutants concentrate and to focus monitoring. The greatest value of this understanding is preventive: the properties of a new chemical can be predicted from similar ones under the precautionary principle, allowing control before release rather than clean-up afterwards.	
	Weaker answer - Level 1 (2 marks)	
	Pollutants have properties like toxicity, persistence and solubility. Toxic pollutants are dangerous. Persistent pollutants like DDT last a long time in the environment and build up in food chains. Soluble pollutants dissolve in water and get into rivers. Knowing these properties helps scientists to predict what pollutants will do and to control them, for example by banning DDT.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 1 defines three properties in isolation; Level 3 pairs every property with a prediction *and* a control strategy, which is what the question asks.	
	• Level 3 uses a wide range of accurate examples (Bhopal, nitrate zones, methyl mercury, PANs, caesium adsorption); Level 1 uses DDT twice.	
	• The final paragraph links properties to environmental conditions (CPA) and to the precautionary principle - synoptic evaluation that distinguishes the top band.	
07.2	1 mark for the point below. **1** The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** Inorganic mercury sank into the sediments / deep water where conditions are anaerobic / low oxygen **1** It was chemically reduced / converted by (anaerobic) bacteria to methyl mercury **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Methyl mercury is liposoluble so it passes through cell membranes and is stored in fat / not excreted **1** So it bioaccumulates in organisms and biomagnifies along the food chain (eg in fish, then humans eating fish) **1** It is a neurotoxin / teratogen causing brain damage and birth abnormalities (eg Minamata) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3

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Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Mercury is persistent / not degraded – it is an element so cannot be broken down **1** It remains in the sediments and continues to be released / converted and taken up by the food chain – long-lived fish continue to accumulate it **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
08.4	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Water current velocity and direction (dispersal) **1 Properties of the pollutant:** solubility, persistence, liposolubility / potential to bioaccumulate **1** Food chain pathways to humans (fish, shellfish) / the critical group **1** Sediment type / adsorption onto clay **1** Volume of the water body / residence time / dilution **1** Sensitive locations downstream (fisheries, drinking water abstraction) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **critical group** → the people most at risk / most heavily exposed • **adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 238-242; AQA spec 3.4.1, 3.4.2.	

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